

Agentic Augmentation for Aspect Based Sentiment Analysis

Mohammad Akbari Monfared
Advisor : Dr. Akbar karimi

Forecast

Problem :

- ABSA needs fine-grained annotations, which are **costly and limited**.
- Using LLMs via **naive prompting results in low-quality, inconsistent outputs**.

Insights :

- **Agentic data generation produces high-quality data.**
- This method can **improve performance**
- the **effectiveness varies** by task, model type, and augmentation size

Outline

- **Motivation**
- **Related Work**
- **Methodology**
- **Results**
- **Summary**
- **Future work**

Motivation

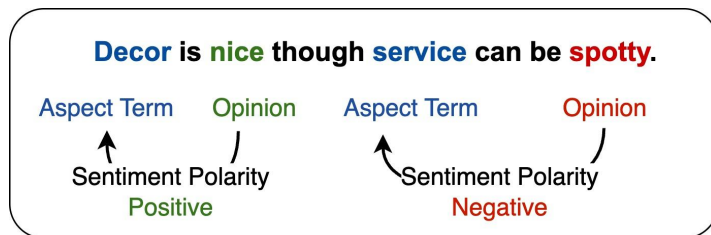
- Huge volume of opinionated text (reviews, social media, feedback).
- **ABSA** (Aspect-Based Sentiment Analysis) → understand sentiment per aspect.

Problem:

- High-quality labeled data is scarce, costly.
- LLMs can generate synthetic data, but often, noisy, or hallucinated.

Need:

- Structured, controlled generation workflows → ensures reliable, label-consistent data.



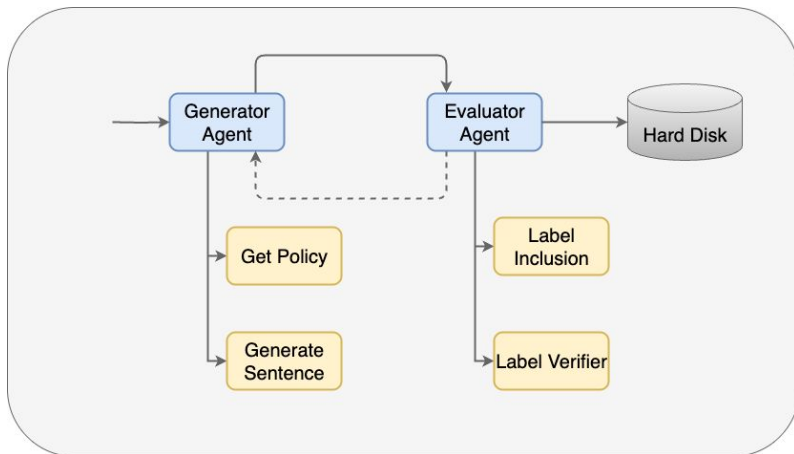
Related Work

- **LLM-based augmentation**
 - ZeroGen, PROGEN → show LLMs can generate synthetic datasets
 - Quality varies: task complexity, prompt design
- **ABSA-specific augmentation**
 - SPM, UDA, IDG → boost performance, but:
 - Usually target **one subtask (ATE / ATSC)**
 - Rely on **prompting or simple filters**
- **Agentic data generation**
 - Recent in NLP: multi-step, self-reflective pipelines
 - Not yet applied to ABSA augmentation
- **Research gaps:**
 1. No **agentic** augmentation system designed for ABSA
 2. No systematic **comparison of agentic vs. prompting** for ABSA
 3. Lack of augmentation strategies spanning **all three subtasks (ATE, ATSC, ASPE)**
 4. Limited study of **model sensitivity** (instruction-tuned vs. base models)

Methodology

Agentic Data Generation Framework for ABSA

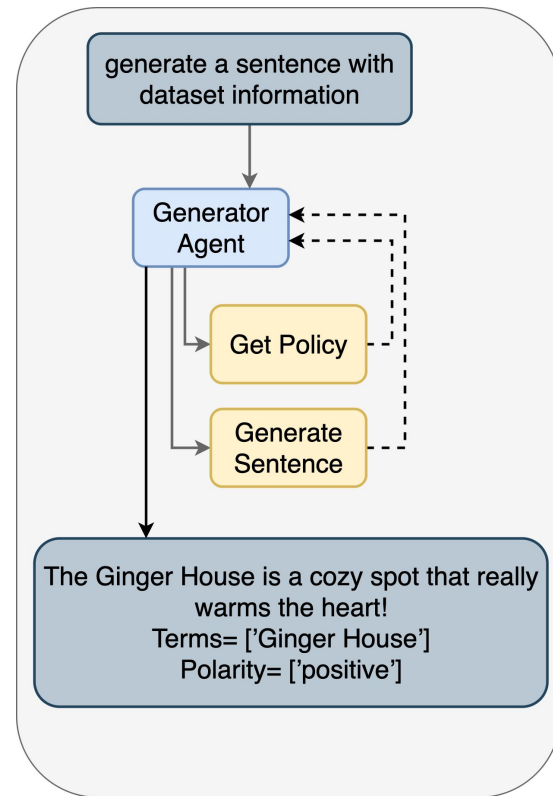
- Agentic system to generate high-quality, structured synthetic data tailored for ABSA
- **ReAct** agents, Implemented using **LangGraph** framework
- Powered by **Qwen 2.5 14B** for agentic brain, and tool use
- Two main agents:
 - **Generator Agent**: Generate potential data points
 - **Evaluator Agent**: Checks the correctness



Methodology

Generator Agent

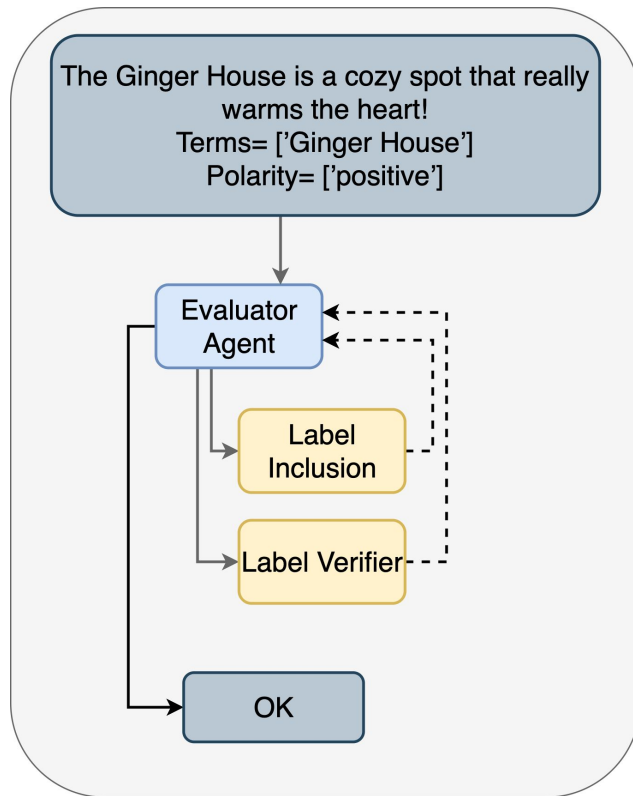
- **Goal:** Generate candidate data points from label
- Given Tools :
 - **Get Policy:** provide information
 - Select labels
 - Sample from label dict, from real dataset
 - Style information
 - Sample 3 sentences from real dataset
 - LLM call to extract Length, Grammar, Writing Style
 - **Generate Sentence:**
 - Generate sentence from info from get policy
 - LLM call with prompt including rules and examples



Methodology

Evaluator Agent

- **Goal:** Ensure label accuracy
- Given Tools :
 - **Label Inclusion:**
 - labels exist in sentence
 - Binary evaluation
 - **Label Verifier:**
 - labels are used as intended term and polarity
 - LLM call with Binary evaluation



Methodology

Raw Prompting

- To **isolate the impact** of the agentic workflow
- same LLM as the agentic system: **Qwen 2.5 14B**.
- Employs the **same label sampling strategy** for aspect term, polarity
- Prompt is **identical** to prompt used in agentic workflow:
 - Examples
 - Generation rules
- No reflection, revision, or quality control — the model's **first response is used**.

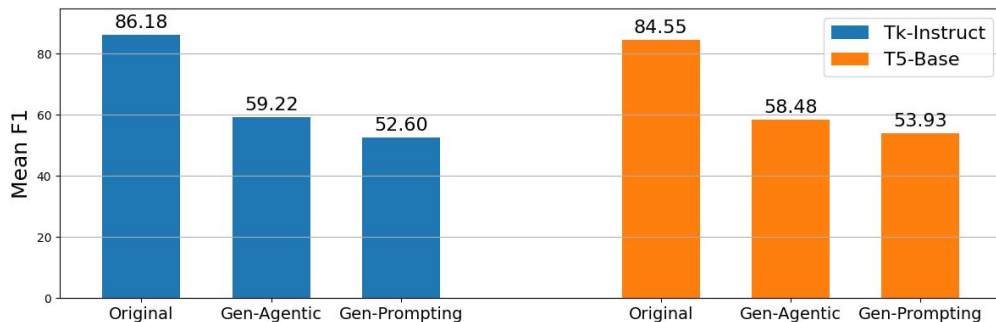
Experiments Setup

- **Tasks:** ATE, ATSC, ASPE
- **Datasets:** SemEval (Laptop 14, Rest14, Rest15, Rest16)
- **Models:**
 - T5-Base : fine tuning
 - Tk-Instruct : InstructAbsa instruction tuning
- **Training Configs:**
 - **Baseline** – Original only
 - **Aug Only** – Synthetic only (Agentic / Prompting, scales: x1, x2)
 - **Train + Aug** – Mixed (Original + Synthetic, scales: x1, x2)
- **Low-Resource:** 30-sample subsets (Original / Agentic / Prompting)
- **Evaluation:** F1 score, avg. of 3 runs

Results

Training on Generated Data Alone

- Large **performance drop** vs. real annotated data
- **Agentic > Prompting** by ~5–6% F1 on average
- Still a **big gap** remains due to:
 - Lower label consistency
 - Reduced sentence complexity



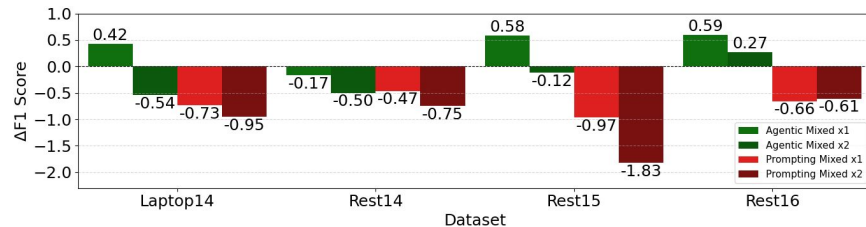
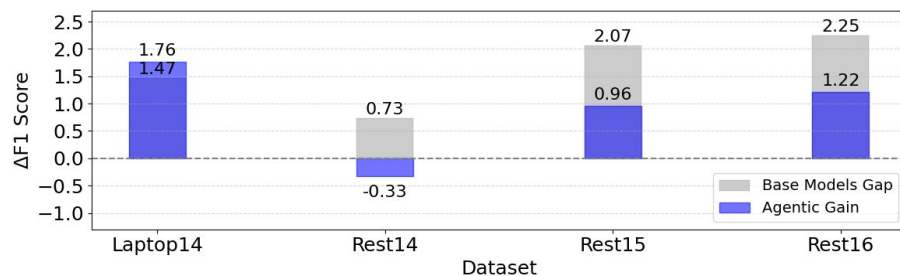
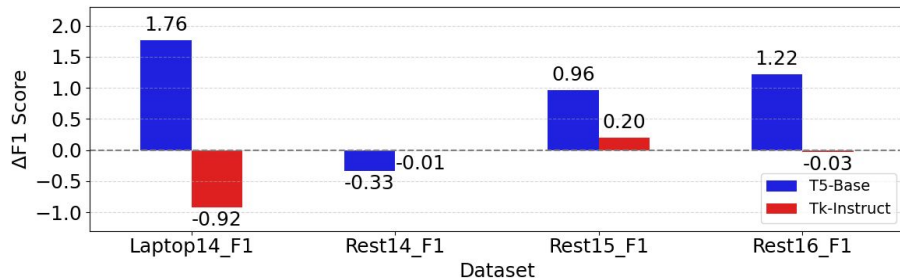
Data Type	ATE	ATSC	ASPE
Agentic	78.17%	80.37%	33.89%
Prompting	43.89%	72.94%	18.33%

Source	Sentence	Aspect Term(s)	Polarity
Laptop14 (Original)	“Reason why? It is because when you buy it, you know first thing that you will not lose any value for that laptop, the price will stay the same for the next year, and even if Apple does decide to change model, your laptop value will only drop 10–20%, unlike PC laptops which drop more than 80%.”	value, price, value	positive, positive, positive
Laptop14 (Original)	“I bought this laptop because of the performance to price ratio.”	performance, price	positive, positive
Agentic (Generated)	“The value is not great, and the specs are underwhelming.”	value, specs	negative, negative
Agentic (Generated)	“The space feels cramped, and the price is way too high for what you get, but at least the frame looks sleek and modern.”	space, price, frame	negative, negative, positive
Prompting (Generated)	“The price is too high, offering poor value for the money.”	price, value	negative, negative

Results

Training on Mixed Data: Model Sensitivity (T5 vs. Tk-Instruct)

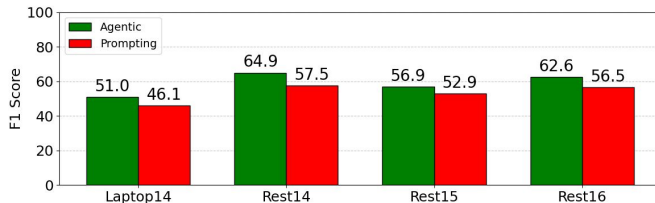
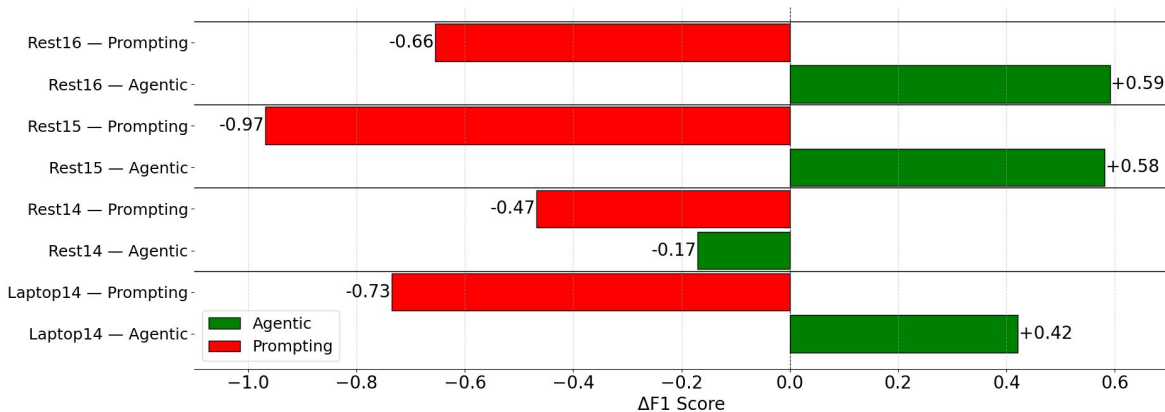
- **T5-Base** benefits more than Tk-Instruct
- Tk-Instruct: **stronger baseline & massive pretraining** = smaller gains
- **Agentic data narrows the gap** across datasets & tasks
- **x2 augmentation often hurts** (too much synthetic data adds noise)



Results

Training on Mixed Data: Agentic vs. Raw Prompting

- **Agentic + Real** → consistent gains
- **Prompting + Real** → little/no gain
- Same **model + prompts**, but different workflows → very different outcomes
- **Consistent trend** due to higher **quality**, evident in training solely in generated data

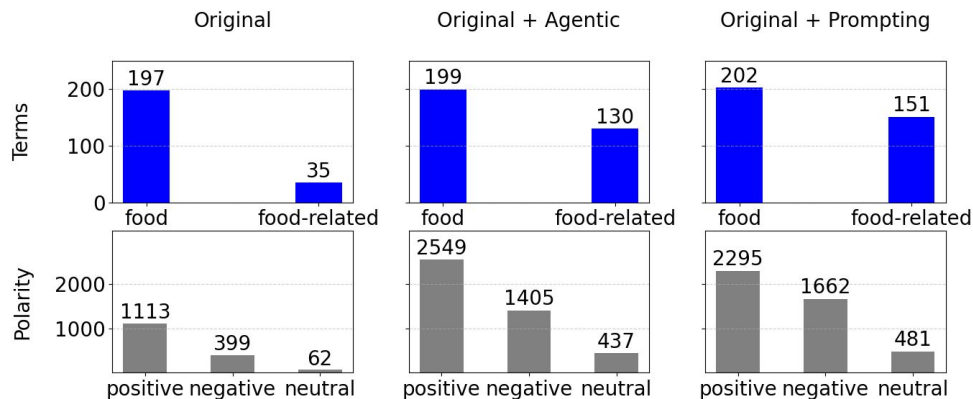


Results

Training on Mixed Data: Agentic vs. Raw Prompting

- **Higher quality :**
 - **Better label consistency** → fewer mislabeled samples
 - **Nuanced, natural sentences** → richer language variety
- **Balance label distributions**

Data Type	ATE	ATSC	ASPE
Agentic	78.17%	80.37%	33.89%
Prompting	43.89%	72.94%	18.33%



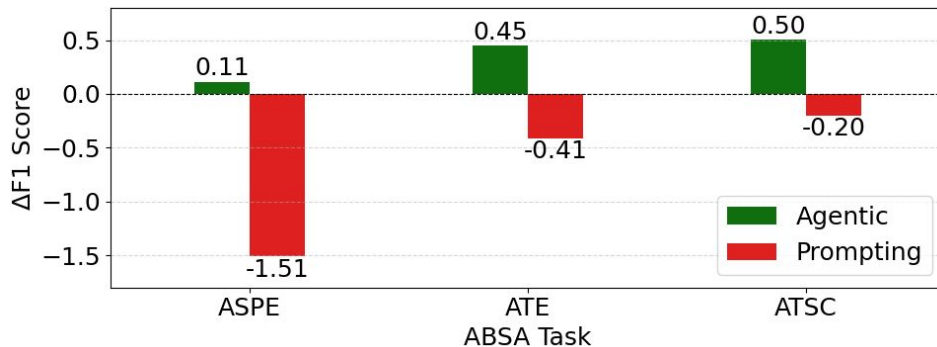
Source	Sentence	Aspect Term(s)	Polarity
Prompting	The chicken was overcooked and dry, while the mashed potatoes were bland and lacked flavor	chicken and mashed potatoes	negative
Prompting	The wait staff at Pizza 33 was friendly and attentive, but the service was slow despite the delicious salmon	Pizza 33, wait staff, salmon	negative, positive, positive
Prompting	The bagels were just okay, but the potato stuff was bland and unappetizing	bagels, potato stuff kanish	neutral, negative
Agentic	Pacifico is a refreshing choice on hot days!	Pacifico	positive
Agentic	The fish and chips were crispy and delicious, the vibe was awesome, and the martinis hit the spot; there was not much to complain about the sitting space though	fish and chips, vibe, martinis, sitting space	positive, positive, positive, neutral
Agentic	That martini was so overpriced and tasted like it came from a bottle instead of being crafted by a bartender, really put a damper on the evening	martini	negative

Results

Training on Mixed Data: Task Sensitivity to Augmentation

- **Impact not uniform** → varies by subtask
- Pattern explained by **task complexity**:
 - **ATSC** = simple classification → biggest & most consistent gains
 - **ATE** = extraction ambiguity → moderate, mixed gains
 - **ASPE** = joint extraction + classification → smallest/unstable gains (esp. Tk-Instruct)

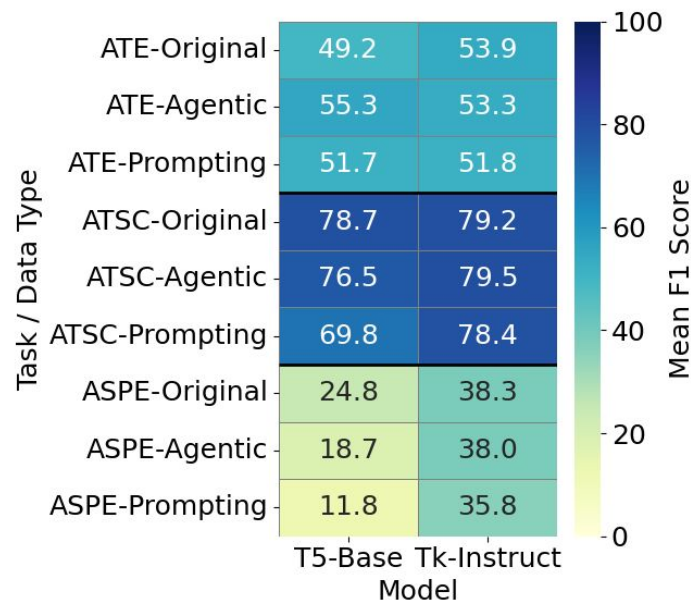
Data Type	ATE	ATSC	ASPE
Agentic	78.17%	80.37%	33.89%
Prompting	43.89%	72.94%	18.33%



Results

Low-Resource Setting (30 Samples)

- **Real > Agentic > Prompting** across tasks
- **Agentic $\approx$ Real** in many cases $\rightarrow$ strong alternative when data scarce
- In ATE (T5-Base), **Agentic > Real**
- **Prompting lags**: poor label consistency & weak structure
- **Task hierarchy remains**: ATSC > ATE > ASPE
- **Tk-Instruct > T5-Base**, especially on complex tasks



Summary

Key Findings

- **Synthetic-only $\neq$ enough** $\rightarrow$ 30–40% F1 drop vs. real data
- **Mixed data helps** $\rightarrow$ esp. for T5 (up to +3% F1), less for Tk-Instruct
- **Agentic $>$ Prompting** $\rightarrow$ +6% F1, higher label consistency & richer sentences
- **T5 + Agentic** can match / beat Tk-Instruct baseline (original-only)
- **Task sensitivity** $\rightarrow$ biggest gains in ATSC, moderate in ATE, minimal in ASPE
- **Low-resource (30 samples)** $\rightarrow$ Agentic $\approx$ Real, Prompting lags

Future Work

- **Task specialized models** → different models for generation vs. validation
- **Model-guided generation** → tailor synthetic data using feedback from training performance
- **Cross-lingual transfer** → extend agentic augmentation to new languages & domains

Questions?



Mohammad Akbari
Monfared

Motivation

Contributions:

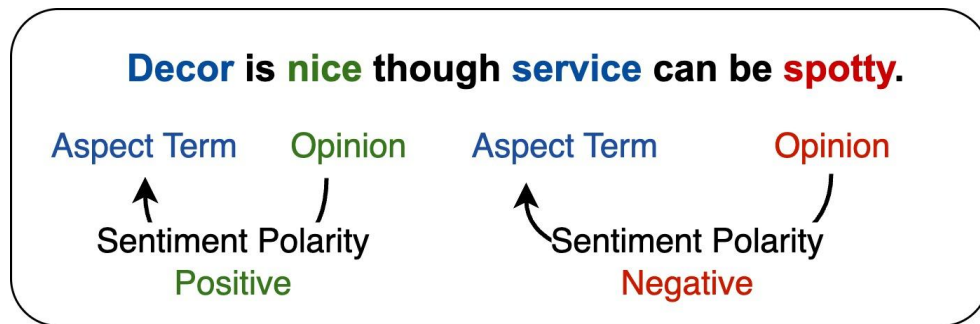
- Introduce an **agentic workflow** for synthetic data tailored to ABSA.
- Develop a **comparable prompting-based method** for fair evaluation.
- Systematically study factors influencing augmentation effectiveness.

Research Questions:

- **RQ1:** Can an agentic workflow improve ABSA model performance by substituting or complementing real annotated data?
- **RQ2:** Can an agentic workflow produce synthetic data that matches or surpasses naive prompting?

Task Definitions

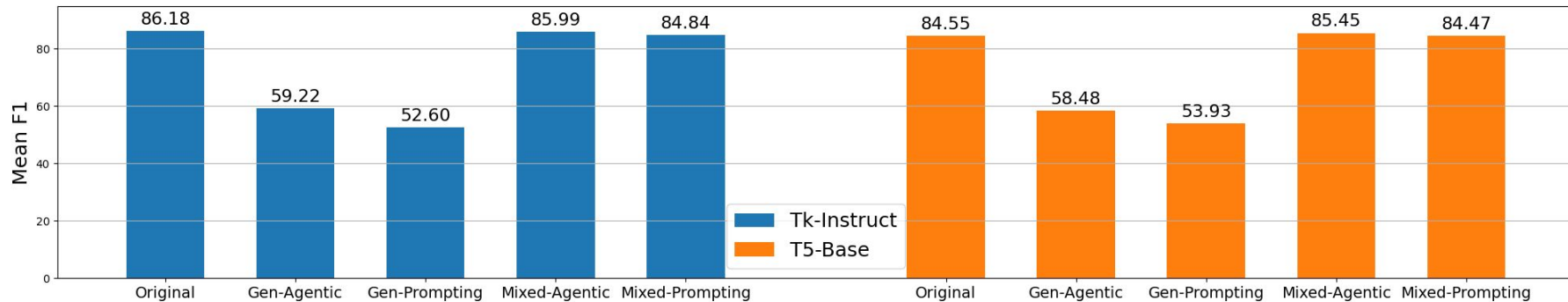
- **ATE :**
 - Given Sentence $\rightarrow$ Aspect Terms : [“Decor” , “Service”]
- **ATSC:**
 - Given Sentence + Aspect Term $\rightarrow$ Polarity : [“Positive”] or [“Negative”]
- **ASPE:**
 - Given Sentence $\rightarrow$ Sentiment Pair : [“Decor: Positive”, “Service: Negative”]



Overall Results

Overall Performance Trends

- **Synthetic-only training** → large performance drop vs. real data
- **Mixed (real + synthetic)** → boosts performance, especially for **T5-Base**
- **Agentic > Prompting** across tasks & models
- **Tk-Instruct** gains are smaller due to strong pretraining



TK-Instruct

From T5 to Tk-Instruct

- **Super-NaturalInstructions dataset**
 - 1,600+ diverse NLP tasks, 5M instances
 - Each task has instructions + positive/negative demonstrations
- **Instruction tuning**
 - Fine-tune T5 on Sup-NatInst tasks
 - Model learns to follow natural language instructions, not just patterns
- **Result: Tk-Instruct**
 - Strong zero-shot generalization to unseen tasks
 - Outperforms larger models like InstructGPT despite smaller size



(a) SUP-NATINST (this work)

Instructions

InstructAbsa Instruction Example

Task	Aspect Term Extraction (ATE)
Definition	Definition: The output will be the aspects (both implicit and explicit) which have an associated opinion that is extracted from the input text. In cases where there are no aspects, the output should be noaspectterm.
Positive Example	Example Input 1: With the great variety on the menu, I eat here often and never get bored. Example Output 1: menu Example Input 2: Great food, good size menu, great service, and an unpretentious setting. Example output 2: food, menu, service, setting
Negative Example	Negative input 1: They did not have mayonnaise, forgot our toast, left out ingredients... Negative output 1: toast, mayonnaise, bacon, ingredients, plate Negative input 2: The seats are uncomfortable if you are sitting against the wall on wooden benches. Negative output 2: seats
Neutral Example	Neutral Input 1: I asked for a seltzer with lime, no ice. Neutral Output 1: seltzer with lime Neutral Input 2: They wouldn't even let me finish my glass of wine before offering another. Neutral Output 2: glass of wine
Input	Now complete the following example- input: My son and his girlfriend both wanted cheeseburgers and they were huge! output: cheeseburgers

Prompt compare

Agentic prompts

Gen Sentence :

You are a critic who can generate comments on the specified aspect and sentiment
We would like you to complete a sentence generation task. Please follow these requirements:

####TASK####

- Generate a sentence using this aspect term: `{', '.join(aspect_term)}` with the following polarities: `{', '.join(polarity)}`.

- Write in the style: `{writing_style}`, and use a `{grammar_structure}` grammatical structure and `{sentence_length}` sentence length.

####REQUIREMENTS####

- Your response must include:

1. The sentence.

2. A line that starts with ``Terms=`` followed by the list of aspect terms used.

3. A line that starts with ``Polarity=`` followed by the matching polarity list.

- Domain: Restaurants

- the sentence should not have aspect words that are not specified in the prompt

- Use the exact structure shown in the examples below.

3 Good Examples , 1 Bad Example

Label verifier :

You are an expert in linguistic evaluation. Your task is to check if the given aspect terms and polarities are correct for the provided sentence.

- If ALL aspect terms appear as actual aspects in the sentence with intended polarities, respond ****only**** with: OK

- If any term is missing, incorrect, or not an aspect of the sentence, or wrong polarities respond ****only**** with: NOT_OK

Do not provide explanations or any other text.

2 Good Example, 2 Bad Examples

Prompt compare

Raw Prompting

You are a critic who can generate comments on the specified aspect and sentiment\n

We would like you to complete a sentence generation task. Please follow these requirements:

- You need to use the sentiment,the aspect mentioned in the prompt\n
- Domain: Restaurants\n
- Your response must include:
 1. The sentence.
 2. A line that starts with `Terms=` followed by the list of aspect terms used.
 3. A line that starts with `Polarity=` followed by the matching polarity list.
- ALL aspect terms appear must as actual aspects in the sentence with intended polarities
- Generated sentence must have the writing style and grammar structure and length of this sentence:{sent}
- the sentence should not have aspect words that are not specified in the prompt\n
- DO NOT REPEAT the input text in the output\n"
- PRINT ONLY THE ANSWER TEXT NO EXPLAINING NOTHING ELSE, MAKE SURE TO USE

ASPECT WORDS IN THE OUTPUT

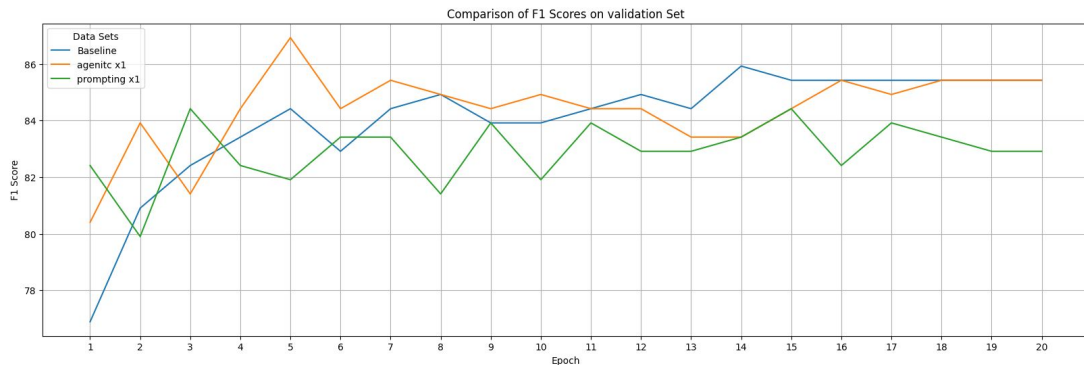
All Examples in Agentic

Now complete this task like example with ONE SENTENCE:

aspect: {combined_aspects} polarity: {combined_polarities}

Training Parameters

- Based on **InstructABSA** baseline for fair comparison
- Only change: **20 epochs**
- Same settings across models and tasks



Models	T5-Base, Tk-Instruct
GPU	1× Nvidia A40
Batch Size	16
Learning Rate	5e-5
Epochs	20

Benchmark Vs. Larger Model

Model	Laptop14	Rest14	Rest15	Rest16
T5-Base	93.32	95.12	80.90	83.28
Tk-Instruct	93.91	95.01	83.41	85.47
Qwen 2.5	60.18	74.37	70.12	72.12

Table 6.2: ATE F_1 baseline scores across models.

Model	Laptop14	Rest14	Rest15	Rest16
T5-Base	79.78	86.25	88.74	91.86
Tk-Instruct	82.75	87.41	88.56	94.30
Qwen 2.5	65.85	78.6	73.62	78.43

Table 6.3: ATSC F_1 baseline scores across models.

Model	Laptop14	Rest14	Rest15	Rest16
T5-Base	81.19	81.34	73.38	79.47
Tk-Instruct	82.03	82.49	77.25	81.60
Qwen	43.8	61.33	58.43	64.07

Table 6.4: ASPE F_1 baseline scores across models.

All Results : mixed

Model	Data Type	Laptop14	Rest14	Rest15	Rest16
T5-Base	Original Data	93.32	95.12	80.90	83.28
	Mixed Data - Agentic				
	Mixed x1	93.82 $\uparrow 0.50$	94.63 $\downarrow 0.49$	80.98 $\uparrow 0.08$	85.21 $\uparrow 1.93$
	Mixed x2	93.50 $\uparrow 0.18$	94.26 $\downarrow 0.86$	82.22 $\uparrow 1.32$	84.37 $\uparrow 1.09$
	Mixed Data - Prompting				
	Mixed x1	92.83 $\downarrow 0.49$	94.53 $\downarrow 0.59$	80.43 $\downarrow 0.47$	83.83 $\uparrow 0.55$
Tk-Instruct	Mixed x2	92.97 $\downarrow 0.35$	94.14 $\downarrow 0.98$	78.88 $\downarrow 2.02$	84.72 $\uparrow 1.44$
	Original Data	93.91	95.01	83.41	85.47
	Mixed Data - Agentic				
	Mixed x1	93.93 $\uparrow 0.02$	94.20 $\downarrow 0.81$	84.12 $\uparrow 0.71$	87.14 $\uparrow 1.67$
	Mixed x2	94.07 $\uparrow 0.16$	93.34 $\downarrow 1.67$	82.50 $\downarrow 0.91$	86.38 $\uparrow 0.91$
	Mixed Data - Prompting				
	Mixed x1	93.80 $\downarrow 0.11$	94.10 $\downarrow 0.91$	82.30 $\downarrow 1.11$	85.32 $\downarrow 0.15$
	Mixed x2	93.35 $\downarrow 0.56$	94.20 $\downarrow 0.81$	80.00 $\downarrow 3.41$	84.39 $\downarrow 1.08$

Table 6.10: ATE F_1 scores across datasets for each model under original and mixed training data. Arrows indicate improvement or drop relative to the baseline.

Model	Data Type	Laptop14	Rest14	Rest15	Rest16
T5-Base	Original Data	79.78	86.25	88.74	91.86
	Mixed Data - Agentic				
	Mixed x1	83.38 $\uparrow 3.60$	86.78 $\uparrow 0.53$	88.92 $\uparrow 0.18$	93.33 $\uparrow 1.47$
	Mixed x2	81.34 $\uparrow 1.56$	86.42 $\uparrow 0.17$	88.56 $\downarrow 0.18$	93.17 $\uparrow 1.31$
	Mixed Data - Prompting				
	Mixed x1	79.93 $\uparrow 0.15$	86.96 $\uparrow 0.71$	88.37 $\downarrow 0.37$	93.82 $\uparrow 1.96$
Tk-Instruct	Mixed x2	81.34 $\uparrow 1.56$	85.62 $\downarrow 0.63$	88.37 $\downarrow 0.37$	93.00 $\uparrow 1.14$
	Original Data	82.75	87.41	88.56	94.30
	Mixed Data - Agentic				
	Mixed x1	80.72 $\downarrow 2.03$	88.30 $\uparrow 0.89$	88.92 $\uparrow 0.36$	93.33 $\downarrow 0.97$
	Mixed x2	81.34 $\downarrow 1.41$	87.41 $\downarrow 0.00$	88.74 $\uparrow 0.18$	92.68 $\downarrow 1.62$
	Mixed Data - Prompting				
	Mixed x1	80.87 $\downarrow 1.88$	87.58 $\uparrow 0.17$	88.00 $\downarrow 0.56$	92.52 $\downarrow 1.78$
	Mixed x2	81.19 $\downarrow 1.56$	87.23 $\downarrow 0.18$	87.60 $\downarrow 0.96$	92.84 $\downarrow 1.46$

Table 6.11: ATSC F_1 scores across datasets for each model under original and mixed training data. Arrows indicate improvement or drop relative to the baseline.

Model	Data Type	Laptop14	Rest14	Rest15	Rest16
T5-Base	Original Data	81.19	81.34	73.38	79.47
	Mixed Data - Agentic				
	Mixed x1	82.37 $\uparrow 1.18$	80.31 $\downarrow 1.03$	76.00 $\uparrow 2.62$	79.72 $\uparrow 0.25$
	Mixed x2	78.84 $\downarrow 2.35$	81.52 $\uparrow 0.18$	73.89 $\uparrow 0.51$	78.79 $\downarrow 0.68$
	Mixed Data - Prompting				
	Mixed x1	80.68 $\downarrow 0.51$	80.42 $\downarrow 0.92$	72.92 $\downarrow 0.46$	79.02 $\downarrow 0.45$
Tk-Instruct	Mixed x2	78.71 $\downarrow 2.48$	80.42 $\downarrow 0.92$	72.34 $\downarrow 1.04$	76.06 $\downarrow 3.41$
	Original Data	82.03	82.49	77.25	81.60
	Mixed Data - Agentic				
	Mixed x1	81.29 $\downarrow 0.74$	82.38 $\downarrow 0.11$	76.79 $\downarrow 0.46$	80.80 $\downarrow 0.80$
	Mixed x2	80.62 $\downarrow 1.41$	81.66 $\downarrow 0.83$	75.60 $\downarrow 1.65$	82.21 $\uparrow 0.61$
	Mixed Data - Prompting				
	Mixed x1	80.46 $\downarrow 1.57$	81.22 $\downarrow 1.27$	74.41 $\downarrow 2.84$	77.54 $\downarrow 4.06$
	Mixed x2	79.70 $\downarrow 2.33$	81.52 $\downarrow 0.97$	74.10 $\downarrow 3.15$	81.30 $\downarrow 0.30$

Table 6.12: ASPE F_1 scores across datasets for each model under original and mixed training data. Arrows indicate improvement or drop relative to the baseline.

All Results : Low Resource Setting

Model	Data Type	Laptop14	Rest14	Rest15	Rest16
T5-Base	Original	41.06	64.68	41.81	49.45
	Agentic	49.44	67.59	50.26	54.09
	Prompting	40.59	68.95	45.42	51.76
Tk-Instruct	Original	44.45	66.24	50.21	54.88
	Agentic	44.1	65.86	47.87	55.18
	Prompting	43.02	64.35	47.68	52.16

Table 6.15: Low-resource ATE F_1 scores across datasets and models.

Model	Data Type	Laptop14	Rest14	Rest15	Rest16
T5-Base	Original	71.31	76.78	78.34	88.38
	Agentic	70.84	74.02	75	86.17
	Prompting	67.22	65.77	73.75	72.58
Tk-Instruct	Original	75.23	76.78	79.15	85.69
	Agentic	74.29	76.87	78.59	88.13
	Prompting	69.12	77.41	78.96	88.13

Table 6.16: Low-resource ATSC F_1 scores across datasets and models.

Model	Data Type	Laptop14	Rest14	Rest15	Rest16
T5-Base	Original	11.43	36.58	25.24	26.06
	Agentic	11.81	25.05	20.94	17.03
	Prompting	3.3	20.34	10.01	13.54
Tk-Instruct	Original	26.31	44.85	39.33	42.52
	Agentic	26.84	46.72	36.42	41.83
	Prompting	24.39	44.4	36.0	38.32

Table 6.17: Low-resource ASPE F_1 scores across datasets and models.